

EE 6984 - Quantum Optics (Assignment 2)

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1. Rabi Oscillations

(a) $\mu_z = -e \langle 2p_z | z | 1s \rangle$

$$\psi_{1s} = \frac{1}{\sqrt{\pi a_0^3}} \exp(-r/a_0), \quad \psi_{2p_z} = \frac{1}{\sqrt{32\pi a_0^3}} \left(\frac{r}{a_0}\right) e^{-r/2a_0} \cos\theta$$

$$z = r \cos\theta, \quad d^3r = r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$\Rightarrow \langle z \rangle = \underbrace{\int_0^{2\pi} d\phi \int_0^\pi \cos^2\theta \sin\theta \, d\theta}_{4\pi/3} \int_0^\infty r^3 \psi_{2p_z}^* \psi_{1s} \, dr$$

radial integral: $\frac{1}{\sqrt{32} \pi a_0^4} \int_0^\infty r^4 e^{-3r/2a_0} \, dr$

we know, $\int_0^\infty x^n e^{-\alpha x} \, dx = \frac{n!}{\alpha^{n+1}}$

$$\Rightarrow \text{radial integral} = \frac{1}{\sqrt{32} \pi a_0^4} \frac{4!}{(-3/2a_0)^5} = \frac{64 a_0}{81 \pi \sqrt{2}}$$

$$\Rightarrow \langle z \rangle = \frac{4\pi}{3} \times \frac{64 a_0}{81 \sqrt{2} \pi} = \frac{128 \sqrt{2}}{243} a_0 \approx 0.745 a_0$$

$$\mu_z \approx e \langle z \rangle = (0.745) e a_0 = \cancel{5.4 \times 5.29 \times 10^{-11} \text{ m}}$$

$$e = 1.602 \times 10^{-19} \text{ C} \quad a_0 = 5.29 \times 10^{-11} \text{ m}$$

$$\boxed{\mu_z \approx 6.31 \times 10^{-30} \text{ C}\cdot\text{m}}$$

$$(b) (i) I = \frac{c \epsilon_0 E_0^2}{2} \Rightarrow E_0 = \sqrt{\frac{2I}{c \epsilon_0}}$$

$$I = 10 \times 10^3 \text{ W m}^{-2} \Rightarrow E_0 \approx 2745 \text{ V/m}$$

$$\Omega = \frac{\mu_2 E}{\hbar} \approx 1.64 \times 10^8 \text{ rad s}^{-1}$$

$$T = 2\pi / \Omega \approx \boxed{38.3 \text{ ns}}$$

$$(ii) T \propto \frac{1}{\Omega} \propto \frac{1}{E} \propto \frac{1}{\sqrt{I}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{I_2}{I_1}}$$

$$\Rightarrow I_2 = \left(\frac{38.3 \text{ ns}}{1.6 \text{ ns}} \right)^2 \times 10^4 \text{ W m}^{-2} \approx \boxed{5.73 \times 10^6 \text{ W m}^{-2}}$$

(2) PULSE AREA

$$(a) \tau_{FWHM} = 1 \text{ ps}, \quad \Theta = \pi/2 \Rightarrow t / \tau_{FWHM} = t \times 10^{12} \text{ s}^{-1}$$

$$E_0(t) = E_{\text{peak}} \exp \left\{ -2 \ln 2 (t \times 10^{12})^2 \right\}$$

$$\int_{-\infty}^{\infty} E_0(t) dt = E_{\text{peak}} \int_{-\infty}^{\infty} \exp \left\{ -2 \ln 2 (t \times 10^{12})^2 \right\} dt$$

$$\text{we know, } \int_{-\infty}^{\infty} e^{-at^2} dt = \sqrt{\frac{\pi}{a}}, \quad a = \frac{2 \ln 2}{(10^{12})^2}$$

$$\Rightarrow \int_{-\infty}^{\infty} E_0(t) dt = E_{\text{peak}} \times 10^{12} \sqrt{\frac{\pi}{2 \ln 2}}$$

$$\Theta = \left| \frac{\mu_2}{\hbar} \int_{-\infty}^{\infty} E_0(t) dt \right| = \frac{\pi}{2} \Rightarrow \frac{\mu_2}{\hbar} E_{\text{peak}} \times 10^{12} \sqrt{\frac{\pi}{2 \ln 2}} = \frac{\pi}{2}$$

$$\Rightarrow E_{\text{peak}}^2 = \frac{\pi \hbar^2 \ln 2}{2 \mu_2^2} \times 10^{24}$$

Fluence, $F = \int_{-\infty}^{\infty} I(t) dt$, $I(t) = \frac{C \epsilon_0 E_0^2(t)}{2}$

$$\Rightarrow F = \frac{1}{2} C \epsilon_0 E_{\text{peak}}^2 \int_{-\infty}^{\infty} e^{-4 \ln 2 (t \times 10^{12})^2} dt$$

$$= \frac{C \epsilon_0 E_{\text{peak}}^2 \times 10^{-12}}{4} \sqrt{\frac{\pi}{\ln 2}}$$

Total pulse energy, $U = \pi r^2 F = \frac{\pi r^2 C \epsilon_0 E_{\text{peak}}^2 \times 10^{-12}}{4} \sqrt{\frac{\pi}{\ln 2}}$

$$\Rightarrow U = \frac{\pi r^2 C \epsilon_0 \times 10^{-12}}{4} \sqrt{\frac{\pi}{\ln 2}} \left(\frac{\pi \hbar^2 \ln 2}{2 \mu_{12}^2 \times 10^{-24}} \right)$$

$$\Rightarrow U = \frac{r^2 C \epsilon_0 \left(\frac{\hbar}{2} \right)^2 \hbar^2 \sqrt{\pi \ln 2}}{2 \mu_{12}^2 \times 10^{-12}} \approx \boxed{5.38 \times 10^{-13} \text{ J}}$$

(b) $|g\rangle = |\psi(0)\rangle$

$$|\psi(t)\rangle = \cos\left(\frac{\Theta}{2}\right) |g\rangle - i \sin\left(\frac{\Theta}{2}\right) |e\rangle$$

$$= \boxed{\frac{1}{\sqrt{2}} (|g\rangle - i |e\rangle)}$$

final state of the system

(3) Density Matrices

(a) Pure state: $\hat{\rho} = |\psi\rangle\langle\psi| \Rightarrow \hat{\rho}^2 = |\psi\rangle(\langle\psi|\psi\rangle)\langle\psi| = \hat{\rho}$

$$\text{Tr}(\hat{\rho}^2) = \text{Tr}(\hat{\rho}) = \sum_i \langle i|\psi\rangle\langle\psi|i\rangle = \sum_i |\langle\psi|i\rangle|^2 = 1$$

Mixed state: $\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|$

expressing $\hat{\rho}$ in its own eigenbasis, it becomes diagonal

$$\hat{\rho} = \sum_n p_n |n\rangle\langle n| \Rightarrow \hat{\rho}^2 = \sum_n p_n^2 |n\rangle\langle n|$$

$$\text{Tr}(\hat{\rho}^2) = \sum_n p_n^2 < \sum_n p_n = 1 \Rightarrow \boxed{\sum_n p_n^2 < 1}$$

$$(b) \langle \hat{O} \rangle = \text{Tr}(\rho \hat{O})$$

$$\begin{aligned} \langle \hat{O} \rangle &= \sum_i p_i \langle \psi_i | \hat{O} | \psi_i \rangle = \sum_{n,i} p_i \langle \psi_i | \hat{O} | n \rangle \langle n | \psi_i \rangle \\ &= \sum_{n,i} \underbrace{\langle \psi_i | \psi_i \rangle}_{\rho_{ii}} \underbrace{\langle \psi_i | \hat{O} | n \rangle}_{\rho_{in}} p_i = \sum_n \langle n | \rho \hat{O} | n \rangle \\ &= \boxed{\text{Tr}(\hat{\rho} \hat{O})} \end{aligned}$$

(c) coherence condition: $|\rho_{ij}| = \sqrt{\rho_{ii} \rho_{jj}}$
that is, it is a pure state.

$$\begin{aligned} (4) a) \rho &= |\psi\rangle\langle\psi| = (a|g\rangle + b|e\rangle)(a^*\langle g| + b^*\langle e|) \\ &= aa^*|g\rangle\langle g| + ab^*|g\rangle\langle e| + ba^*|e\rangle\langle g| + bb^*|e\rangle\langle e| \end{aligned}$$

$$\Rightarrow \boxed{\rho = \frac{3}{4}|g\rangle\langle g| + \frac{i\sqrt{3}}{4}|e\rangle\langle g| - \frac{i\sqrt{3}}{4}|g\rangle\langle e| + \frac{1}{4}|e\rangle\langle e|}$$

$$b) \left\{ \frac{3}{4}, |g\rangle \right\}, \left\{ \frac{1}{4}, |e\rangle \right\}$$

$$\Rightarrow \rho = \begin{bmatrix} 3/4 & 0 \\ 0 & 1/4 \end{bmatrix}$$

c) population measurement will give exactly same results, 75% in $|g\rangle$ and 25% in $|e\rangle$. Hence, we'd have to probe the off-diagonal elements.

Apply $\hat{O} = \hat{\sigma}_y$ and measure.

$$\text{For } \rho_{\text{pure}}, \text{Tr}(\hat{\rho} \hat{O}) = \sqrt{3}/2$$

$$\text{For } \rho_{\text{mixed}}, \text{Tr}(\hat{\rho} \hat{O}) = 0$$

Hence measuring the dipole moment can be used for distinguishing.

